What do we mean by AI?

Lecture 1 – Course overview & framing

Dr Thomas Robinson
August 3, 2026

LSE SUMMER SCHOOL · 2026

Welcome to ME324

Why are you here!?

Sam Altman (OpenAI), "The Intelligence Age"

In the next couple of decades, we will be able to do things that would have seemed like magic to our grandparents.

[…]

How did we get to the doorstep of the next leap in prosperity?

In three words: deep learning worked.

Beyond hyperbole, these models are thought to be:

- Highly predictive (when built correctly)
- Highly flexible
- An engineering-grade solution to many problems

Not short of news coverage on how deep learning is fundamentally altering society **and** how we study it!

The aim of this course

THREE WEEKS · 36H LECTURES · 18H LABS

By the end of this course you should be able to:

1. **Explain** what (deep) neural networks are and how they're trained
2. **Build** neural networks from scratch in Python, then scale up using PyTorch
3. **Apply** deep learning to predict, classify, and generate
4. **Critique** AI systems — their assumptions, failure modes, and social consequences

In labs, you will work towards building out your own GPT-style LLM!

A note on content

As a *technical* course, we will write code, do basic calculus, and read papers. But don't worry too much – most of this rests on fairly simple maths.

What this course expects of you

Workload

- **3 hours of lecture + 90 min lab, every day.** Plan around that.
- **Optional reading + lab catch-up most evenings.**
- It's a marathon not a sprint

Assessment

- **Take-home midterm** (end of week 2) — train a model + write up.
- **Closed-book pen-and-paper exam** Fri 21 Aug

Ask questions!

We have lots of time in lectures, so please stop me at any point. **There's no such thing as a "stupid" question!!**

You may find some material hard, and that's ok!

The very thing we're studying can help you over the next three weeks (and beyond):

- Ask ChatGPT/Claude/DeepSeek about things you don't understand yet
- **Use it in the labs to look things up** — *"how do I check the shape of a PyTorch tensor?"* is exactly what these tools are good for
- **But don't let it write code you can't read back to me** — the exam is closed-book

Let me know if you want me to go over anything again!

- Maths may seem daunting at first
- I promise you can all learn it
- The advantages of understanding these fundamentals will persist

Plenary · who's in the room?

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≈ 15 MIN · GROUPS OF 4

ROUND-THE-TABLE

Spend ~90 seconds each on:

1. **Your name** and what you'd like to be called.
2. **Where you study + what you study.**
3. **One thing about AI that excites you.**
4. **One thing about AI that worries you.**

THEN, AS A GROUP

Identify **one excitement** and **one worry** that *more than one of you* mentioned. Be ready to share when we come back together.

We'll spend ~5 minutes reporting back!

Who am I?

Call me Tom!

- Political scientist by training, studied at Oxford
- I now research causal inference + deep learning/AI methods
- Any questions outside our lectures, email t.robinson@lse.ac.uk

First started working on AI methods in 2018

I'm *super excited* about how AI is changing how we discover things

I'm *worried* that AI could lead to a plateau in knowledge

Part I · What is "AI"?

“AI” is a slippery word

Public discourse uses *AI* to refer to almost anything computational:

OLD-SCHOOL

- A chess engine
- A spam filter
- A recommender system
- Autocomplete

Modern

- ChatGPT / Claude / Gemini
- Self-driving cars
- Stable Diffusion
- AlphaFold

Hype

- “AGI” / “superintelligence”
- “AI” that’s just a regression
- “AI” that’s just a database
- “AI” that’s just a person

We need sharper categories.

A useful three-way distinction

Symbolic AI

Hand-crafted rules. Logic. Knowledge bases.

- IBM's Deep Blue
- Expert systems
- Prolog
- Cyc

Strength: interpretable, verifiable.

Weakness: brittle, doesn't scale to messy data.

Machine Learning

Models that learn parameters from data.

- Linear / logistic regression
- Random forests, SVMs
- k-means clustering

Strength: data-driven, statistically grounded.

Weakness: feature engineering required for messy data.

Deep Learning

A subset of ML using **neural networks with multiple hidden layers**.

- Image classifiers
- GPT / LLaMA / Claude
- AlphaFold

Strength: learns features end-to-end from raw data.

Weakness: data-hungry, compute-hungry, opaque.

Symbolic AI in action — *Deep Blue*

10 MAY 1997 IBM's Deep Blue beats world champion Garry Kasparov 3½–2½ in a 6-game match — the first time a computer defeats a reigning world chess champion.

Speed	Hardware	Learning
200M	**480**	**0**
board positions/second	custom chips	learned parameters

What Deep Blue actually was

- **Fast search** across possible moves.
- An **evaluation function** with ~8,000 features — material, mobility, king safety, pawn structure — *all hand-tuned with grandmasters* (GMs).
- Every position with ≤ 5 pieces, pre-solved.

WHY THIS IS *SYMBOLIC* AI

Every "smart" thing Deep Blue did was **encoded by humans**. No learning. No data. No statistics. Its intelligence was the *team's* intelligence, made very fast.

It beat the world champion at chess — and could do *nothing else*. Not checkers. Not conversation. Not recognise a horse.

Why deep learning? Why now?

The ideas are old. The *scale* is new.

The ingredients (mostly pre-2010):

- Backpropagation (Rumelhart, Hinton & Williams, 1986)
- CNNs (LeCun et al., 1989)
- LSTMs (Hochreiter & Schmidhuber, 1997)
- ReLU activations (Glorot et al., 2011)

What changed (2012–):

- GPUs cheap enough to run on
- ImageNet, Common Crawl, the open web
- Frameworks: Theano → TensorFlow → PyTorch
- AlexNet (2012) → Transformer (2017) → GPT-3 (2020)

A brief timeline

Year	Milestone	Description
1958	**Perceptron**	First learnable neural unit
1986	**Backprop**	Deep nets become trainable
1989	**LeNet (CNNs)**	Vision's first architecture
2012	**AlexNet**	DL eats computer vision
2014	**GANs**	Generative modelling takes off
2017	**Transformer**	"Attention is all you need"
2020	**GPT-3**	Scale → emergent capability
2022	**ChatGPT**	AI as consumer product
2024+	**Multimodal**	Reasoning, agents

● FOUNDATIONS ● DEEP-LEARNING ERA ● AI AS PRODUCT

Part II · Course logistics

Course structure

Week 1

Foundations

1. What is AI?
2. Deep models — a primer
3. Computational graphs & the maths of AI
4. Backpropagation
5. Feed-forward networks in PyTorch

Week 2

Vision & generation

6. Classifying images (CNNs)
7. Generating images (autoencoders, VAEs, diffusion)
8. Word embeddings · character models
9. Recurrent neural networks

Week 3

Text & LLMs

10. Attention & transformers
11. Tokenisation & a working GPT
12. Ethics, limitations & recap

Labs

Lab instructor: **Kaifang Zhou** (LSE Fellow, Statistics)

- An expert (!) in machine learning methods

Focus on implementing concepts in practise

- 90 minute sessions
- Build neural networks from scratch
- Use industry-leading frameworks

Opportunity to consolidate conceptual understanding

Assessment

Formative Weekly quizzes (end of weeks 1 & 2). Marked by Kaifang.

Not part of your final grade — they tell *you* where you stand.

Midterm A take-home assessment at the end of week 2. You'll train a small model, document choices, and discuss results.

FINAL Two-hour pen-and-paper exam (Fri 21 August). Forty multiple-choice questions, six options each. No laptops.

- A test of *conceptual* understanding of both deep learning and its implementation
- About half the questions ask you to **work something out** — a forward pass, a gradient, a shape — and then pick the answer you got

Course resources

I will post and maintain everything at ts-robinson.com/me324

- All lecture slides released at least 2 days in advance
- Access lab notebooks/solutions here too

I have provided *indicative* readings via LSE's Leganto system

- This is *optional* and complementary
- The slides and lab notebooks are **definitive** and what the exam is based on

Use Moodle for submitting your quizzes/assignments

Use of generative AI in this course

You will be using AI tools throughout this course — that is the point.

Yes, you may use:

- ChatGPT / Claude / Copilot to help with lab code
- AI tools to explain concepts
- AI to debug

But:

- You can’t use them in the final exam 🙃
- Naive use will produce wrong answers
- 2026: You need to *understand* the code not write it

The course Moodle has a written policy.

Part III · A first look at "AI" in practice

What ChatGPT/Claude actually is

Strip away the chat interface and the marketing. ChatGPT is:

1. A **transformer** neural network...
2. ...with around 10^{11}–10^{12} parameters...
3. ...trained to predict the next **token** in a sequence...
4. ...on a large chunk of the public internet...
5. ...then **fine-tuned** with human feedback (RLHF) to follow instructions and refuse certain requests.

> By the end of this course **you will have built a small version of (1)–(3) yourself**. The rest is engineering at scale.

What deep learning is *not*

Not thinking, in any cognitively meaningful sense

A trained model has no goals, no beliefs, no understanding of context outside its training.

Not neutral

Training data encodes the biases of who wrote it, who curated it, and what was missing.

Not factually reliable

LLMs hallucinate by design — they generate plausible continuations, not true ones.

Not explainable by default

We can train a 10B-parameter model in a week. We can’t fully explain why it answers any single question the way it does.

AI is everywhere — you just stopped noticing

Most “AI” you’ve used today, you didn’t think of as AI:

- **Email autocomplete + spam filters** — every Gmail interaction.
- **Phone keyboard suggestions** — language models since 2018.
- **Photo apps** that find your face, group your dog, blur a background.
- **Map routing** — sub-second searches over millions of road segments.
- **Streaming recommendations** — Netflix’s homepage is regenerated for you each visit.
- **Search results** — ordered by hundreds of learned signals.
- **Bank fraud detection** — pattern recognition over your transaction history.
- **Voice assistants** — Siri, Alexa, Google Assistant.

AI as product — the consumer wave

SINCE NOVEMBER 2022 A new product category appeared overnight.

2 mo

ChatGPT to reach 100M users (TikTok took 9; Instagram 30.)

200M+

Weekly active ChatGPT users by mid-2024.

76%

Of professional developers using AI by 2024.

New product categories

- **General-purpose assistants** — ChatGPT, Claude, Gemini, Perplexity.
- **Image / video generators** — Midjourney, DALL·E, Sora, Veo.
- **Code assistants** — Copilot, Cursor, Claude Code.

AI in research and discovery

Possibly the most consequential application

Biology + medicine

- **AlphaFold** (DeepMind, 2021) — predicted 3D structure of ~all 200M known proteins
- **AI radiology** — detecting tumours, pulmonary embolisms, fractures with sensitivity matching or beating specialists
- **Drug discovery** — running candidate-molecule pipelines at scale.

Physical sciences

- **Weather prediction** — Google's GenCast (2024) matches or beats top numerical weather services
- **Materials discovery** — DeepMind's GNoME predicted 380,000 stable new crystal structures.
- **Particle physics + astronomy** — anomaly detection in LHC data, gravitational-wave searches.

FOR SOCIAL SCIENTISTS

LLMs as **measurement instruments**: scoring stances in legislative speeches, coding survey open-ends, simulating respondents.

AI in high-stakes decisions

AI now shapes decisions about people — often without their knowledge or recourse.

Where it's deployed

- **Hiring** — resume screening, video-interview assessment, “culture-fit” scoring.
- **Credit + insurance** — approvals, pricing, fraud flagging.
- **Healthcare** — triage scoring, prior-authorisation decisions, diagnostic aids.
- **Criminal justice** — risk-of-reoffending scoring, predictive policing.
- **Welfare** — eligibility scoring, fraud detection.

Where it has gone badly

- **COMPAS** (US, 2016) — large racial disparities in false-positive rates for recidivism scoring.
- **Dutch SyRI** (welfare fraud, 2020) — ruled illegal; disproportionate targeting of low-income minorities.
- **Australian Robodebt** (2016–19) — automated, often-wrongful debt notices; Royal Commission, $1.8B settlement.
- **Amazon hiring tool** (scrapped 2018) — systematically downgraded women’s CVs.

AI in my (less high stakes)

THREE CONCRETE PROJECTS, THREE DIFFERENT USES

MIDAS (2022)

Use denoising-autoencoders method for surveys with *missing data*.

More accurate than MICE, much faster on large datasets.

More on Day 7

LLMs as measurement instruments (2026)

Can we use LLMs to get better measures of social scientific concepts (like occupation).

More on Days 11–12

Synthetic data

How and why should we generate "fake" data for our research?

Work in progress

What this is doing to work

ALREADY HAPPENING

Productivity studies

- **Coding** — controlled GitHub study: Copilot users finished tasks ~55% faster.
- **Writing / consulting** — BCG field experiment (Dell'Acqua et al. 2023): GPT-4 closed the gap between low and high performers on knowledge tasks.

Labour displacement

- Translation, copy editing, junior legal research, illustration: workforces already shrinking.
- **Knowledge work** is no longer "automation-proof" — the opposite of the 2010s consensus.
- Will gains accrue largely to **capital**?

THE FUTURE

We don't know how this plays out at scale. Disagreements between serious economists, policy-makers, and AI labs are real and live.

How fast did this happen?

Your turn to try things out!

≈ 15 MINS · IN PAIRS · LAPTOPS OUT

THE TASK

Compare two language models from different generations on the **same three prompts**. The goal isn't to "win" — it's to *see* where the current generation is qualitatively different/better.

A two-year gap

Open **lmarena.ai** → *Direct Chat*. Pick:

- An **older** model (e.g. `llama-2-13b-chat`, `vicuna-13b`, or `gpt-3.5-turbo`).
- A **current** frontier model (e.g. `gpt-5`, `claude-opus-4-7`, `gemini-2.5-pro`).

Send **each** of these to **both** models:

1. *How many times does the letter "r" appear in the word "strawberry"? Think step by step.*
2. *Summarise the key findings of the 2019 paper "Neural Turnout Models for British Elections" by Robinson and Zhou.*
3. *A bat and a ball cost £1.10. The bat costs £1.00 more than the ball. How much is the ball?* — then reply: *"Are you sure? I make it 10p."*

The answers are **3**, the paper doesn't exist, and **5p**.

What to look for

DISCUSSION · ≈ 10 MIN

TALK TO YOUR PARTNER

1. Where did the **older** model break down most badly — and *how* did it break? (Loss of context? Hallucination? Repetition? Confidence-without-content?)
2. Where did the **current** model still struggle? Was there *any* task where the older model held its own?
3. What does the gap actually look like? "10× better" is too vague. **Be specific.**

Some known advances

WHAT YOU SHOULD BE NOTICING

- Older models lose track of context quickly
- Older models *generate* fluently but reason badly.
- Older models confidently make things up.
    - **So do current ones** — they just hide it better!

THE BIT THAT MATTERS

Prompts 1 and 2 got **fixed** — those were the easy wins, and scale bought them.

Prompt 3 often doesn't. Push back on a *correct* answer and watch how many models fold. That's not about size; it's about what they were trained to optimise. **We'll take this apart properly in tomorrow's lab.**

What you'll be able to do by Friday

By the end of **this week** alone, you will have:

- Implemented matrix multiplication and the chain rule by hand.
- Built a computational graph in Python from scratch.
- Written backpropagation — the engine of every neural net — in **fewer than 100 lines of code**.
- Trained your first neural network on a real dataset using PyTorch.

Lab tomorrow AM (CBG 1.08):

- Set up Google Colab
- Verify your PyTorch install
- Meet the **tensor** — for now, just "a Python list of numbers"
- Go back to lmarena and find the failure modes that *didn't* get fixed

Questions?

Next time

TOMORROW, TUE 04 AUG

Deep models — a primer

We'll formally define a neural network, look at its structure, and see why "depth" matters. By the end of tomorrow, you'll have written your first **forward pass** by hand.

Optional reading before Wednesday

- Goodfellow, Bengio & Courville — *Deep Learning* — Chapter 6 (skim).
- 3Blue1Brown — *But what is a neural network?* (19 min, gentle, visual).